

DATA STRUCTURES & ALGORITHMS LAB

PRACTICAL FILE



SCHOOL OF COMPUTER SCIENCE

UNIVERSITY OF PETROLEUM AND ENERGY STUDIES
DEHRADUN (INDIA), 248007

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INDEX:

S.No.	Objective	Date	Sign.
1.	1. Maximum Subarray Sum - Kadane's Algorithm Given an integer array arr[], find the subarray (containing at least one element) which has the maximum possible sum, and return that sum 2. Maximum consecutive ones after flipping zeroes • Input Array • Constant value=k (the no. of zeroes you can flip)	15/08/25	
2.	1. Create a linked list by taking input from the user. 2. Find the middle of the linked list without using the length function. 3. Perform the deletion operation at the beginning of the linked list	03/09/25	
3.	Reverse a linked list • Entire linked list • Reversing the linked list using given positions of start and stop.	03/09/25	
4.	1. Given a string s containing three types of brackets {}, () and []. Determine whether the Expression are balanced or not. An expression is balanced if each opening bracket has a corresponding closing bracket of the same type, the pairs are properly ordered and no bracket closes before its matching opening bracket. 2. Given a stack with push(), pop(), and empty() operations, The task is to delete the middle element of it without using any additional data structure. 3. Implement Stack operations using Linked List.	24/09/25	
5.	1. Implement queue operations using Array. 2. Implement queue operations using Linked list. 3. Implement circular queue using array and linked list	27/09/25	
6.	1. Implement Quick sort algorithm. Find execution time for already sorted order input and unsorted input.	01/10/25	

7.	2. Implement Linear search algorithm. 3. Implement binary search algorithm.	01/10/25	
8.	1. Represent tree data structure with array 2. Represent Tree data structure with linked list 3. apply pre, in, post, level order traversal to display 4. search an element from a binary tree	29/10/25	
9.	Q1. Take random inputs from user and build a Max heap. Q2. Implement the concept of Heap sort algorithm.	19/11/25	